

Technical Verification Companion: Λ CDM+ ΔN_{eff} MCMC Proxy, NaMaster Pipeline Recovery, and Spectator-ALP Consistency Check for the ECH Spin-Torsion Program

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We report the technical verification material for the Einstein-Cartan-Holst (ECH) spin-torsion cosmology no-go program of Paper I(a) [?]. Three analyses are documented. (1) *Stock-CAMB Λ CDM+ ΔN_{eff} MCMC proxy* (Cobaya v3.6.1, 424,781 samples across three dataset combinations): this run uses stock CAMB with ΔN_{eff} as a free parameter and carries *no torsion modifications to the Boltzmann equations*; it is reported as a null-consistency test of an extra radiation-like degree of freedom, not as evidence for or against the ECH spin-torsion framework. Both frozen dataset combinations find ΔN_{eff} consistent with zero (-0.020 ± 0.169 full-tension; $+0.065 \pm 0.17$ Planck+BAO+SN) and $H_0 = 67.68 \pm 1.06 \text{ km s}^{-1} \text{ Mpc}^{-1}$, recovering standard Λ CDM. (2) *NaMaster pseudo- C_ℓ pipeline validation* on the Planck Commander CMB polarization map ($N_{\text{side}} = 512$, $\ell_{\text{max}} = 1024$, $f_{\text{sky}} = 0.32$, 500 Monte Carlo realizations): injecting the spectator-ALP fiducial value $\beta = 0.27^\circ$ recovers $\hat{\beta} = 0.238^\circ$ (pipeline-recovery bias 0.032° , SNR= 20.32). The primary sky detection significance is the published Planck/ACT DR6 2.4–2.9 σ [1, 2]; the pipeline SNR figures refer to recovery of injected MC signals and are *not* competitive sky measurements. (3) *Spectator-ALP consistency check*: a field with $f_a \sim M_{\text{Pl}}$, $m \sim H_0$ is consistent with the published joint Planck+ACT value $\beta = 0.342^\circ \pm 0.094^\circ$ (3.6 σ) [?] without fine-tuning. The same birefringence arises in standard GR with an identical ALP; it is not a distinctive ECH prediction. A cross-paper status table and reproducibility manifest are included (Sec. VII).

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I. INTRODUCTION

This companion paper provides the technical verification layer for the ECH structural-closure no-go result reported in Paper I(a) [?]. The main paper establishes 14 independent structural constraints closing minimal-ECH dark-energy routes and proves a perturbation-transparency theorem for the Holst sector. The present paper documents the three numerical analyses that support and contextualize those results.

Scope of this paper.—Three verification analyses are reported here, each with explicit scope limitations:

1. Λ CDM+ ΔN_{eff} MCMC proxy (Secs. II and III): stock CAMB Boltzmann solver with ΔN_{eff} as a free parameter. **Not a spin-torsion theory module.** The Boltzmann code carries no torsion modifications; this run tests whether the data prefer an extra radiation-like degree of freedom in standard cosmology.
2. NaMaster CMB E - B analysis (Sec. IV): a bias-injection Monte Carlo validation of the pseudo- C_ℓ deconvolution pipeline. **Not a competitive sky detection.** The high pipeline-recovery SNR figures (e.g., 20.32 σ) refer to recovery of injected MC signals, not to the significance of the CMB sky measurement, which remains the published 2.4–2.9 σ from Planck/ACT.

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3. *Spectator-ALP consistency check* (Sec. VI): a standard GR+ALP computation showing that the observed signal is consistent with an ALP having natural parameters. **Not a distinctive ECH prediction.** The same $\beta \approx 0.27^\circ$ arises in any GR+ALP setup with the same parameters; no ECH-specific derivation connects the Holst action to the photon-torsion coupling required.

What is NOT in this paper.—The 14 structural barriers, the perturbation-transparency theorem, the 14-barrier table, and the surviving mechanism-independent test predictions ($f_{\text{NL}} = -35/8$ and ALP birefringence) are in Paper I(a) [?]. The SPHEREx multi-tracer Fisher forecast is in Paper II [?]. The multi-survey anomaly catalog is in Paper III [?]. The galaxy chirality catalog is in Paper IV [?].

Cross-paper citations.—When this companion reports MCMC values (H_0 , σ_8 , etc.) that are referenced in the main paper, those values come from Secs. III and V here.

II. COSMOLOGICAL TENSIONS: H_0 AND σ_8

The bounce scenario motivates extending ΛCDM by ΔN_{eff} (particle production at the bounce) and $(\omega/H)_0$ (angular momentum transfer), treated as phenomenological parameters. Ω_k is fixed to zero (mandated by 92 e -folds of post-bounce inflation). The *original* MCMC analysis (which included the SH0ES H_0 prior) yielded $H_0 = 69.2 \pm 0.8$, $\sigma_8 = 0.785 \pm 0.016$; however, independent verification (Sec. III) shows these were driven by the SH0ES prior, not by the ΔN_{eff} extension. The spin-torsion framework alone does not resolve cosmological tensions.

III. STOCK-CAMB $\Lambda\text{CDM}+\Delta N_{\text{eff}}$ MCMC: GENERIC RADIATION-PROXY TEST (NOT A SPIN-TORSION THEORY MODULE)

Scope statement.—“MCMC verification” refers throughout to a stock CAMB run of ΛCDM extended by ΔN_{eff} as a free parameter. *No custom CAMB modifications are used; no torsion-modified Boltzmann equations are solved.* The MCMC therefore tests whether the data prefer an extra radiation-like degree of freedom, treated as a generic phenomenological proxy for the spin-torsion sector’s possible effective radiation contribution. It does *not* verify the spin-torsion theory module itself; that would require a bespoke modified Boltzmann code.

The proxy run (Cobaya v3.6.1 with Planck NPIPE CamSpec TTTEEE + lowl TT/EE + lensing) has produced two frozen dataset combinations with publication-quality convergence, plus an ongoing Planck-only run (114,992 raw samples). Total MCMC program: 424,781 raw samples across 3 dataset combinations. We report

TABLE I. $\Lambda\text{CDM}+\Delta N_{\text{eff}}$ proxy MCMC results from frozen Cobaya v3.6.1 chains (CAMB v1.6.5, stock; *no torsion modifications*). All values are posterior means $\pm 1\sigma$. Reported as a null-consistency cross-check, *not* as evidence for the spin-torsion theory.

Parameter	Full-tension	Planck+BAO+SN
H_0 [km/s/Mpc]	67.68 ± 1.06	67.79 ± 1.09
ΔN_{eff}	-0.020 ± 0.169	$+0.065 \pm 0.17$
σ_8	0.803 ± 0.008	0.812 ± 0.009
S_8	0.814 ± 0.008	0.831 ± 0.018
Ω_m	0.308 ± 0.005	0.312 ± 0.006
τ	0.054 ± 0.007	0.056 ± 0.007
n_s	0.965 ± 0.004	0.967 ± 0.006
Chains	6	6
Total samples	176,840	132,949
Worst $\hat{R} - 1^a$	0.001	0.003
Min ESS	4,744	4,692

^a Sourced from `convergence_latest.csv`: worst row is n_s in the full-tension combination at $\hat{R} - 1 = 9.74 \times 10^{-4}$; all 14 sampled parameters across both frozen combinations satisfy $\hat{R} - 1 < 3 \times 10^{-3}$. *Not* the stale mid-burn-in diagnostic `convergence_gpu_20260305_stale.csv` ($\hat{R} - 1 \in [0.23, 0.86]$), preserved as a transparency artifact only.

this as a $\Lambda\text{CDM}+\Delta N_{\text{eff}}$ null-consistency test: the data are consistent with $\Delta N_{\text{eff}} = 0$ in stock CAMB.¹

Key finding.—Both frozen datasets find ΔN_{eff} consistent with zero and H_0 consistent with Planck ΛCDM at 0.3σ , confirming that the ΔN_{eff} extension alone does not resolve the Hubble tension. Current data neither require nor exclude a small positive ΔN_{eff} from the spin-torsion sector; CMB-S4 ($\sigma(N_{\text{eff}}) \sim 0.03$) will provide the first precision test.

Independent cross-validation.—Liu *et al.* [3] constrained an EC torsion model using DESI DR2 + PantheonPlus + DES Y5 + Planck 2018, finding torsion preferred by AIC ($\Delta\text{AIC} = -5.7$ to -6.6). Our MCMC agrees at 0.5σ in H_0 and 0.4σ in σ_8 .

IV. DATA METHODS: CMB E - B ANALYSIS

Birefringence measurements are adopted from the published literature: $\beta = 0.30^\circ \pm 0.11^\circ$ (Planck NPIPE [1]) and $\beta = 0.215^\circ \pm 0.074^\circ$ (ACT DR6 [2]). The spectator ALP analysis (Sec. VI) uses these published values.

Scope note.—*The NaMaster pseudo- C_ℓ analysis below is a bias-injection Monte Carlo validation of the decon-*

¹ Sample-count stratification: **424,781** (abstract) is the total raw-accepted count across all three combinations. **309,789** is the sum of the two frozen combinations only (176,840 + 132,949). After removing the first 30% of each chain as burn-in, **123,129** post-burnin samples remain (`convergence_summary.json`); the **119,617** figure in Fig. 1 reflects additional `getdist` weight-based thinning. All four numbers correspond to the same underlying chains.

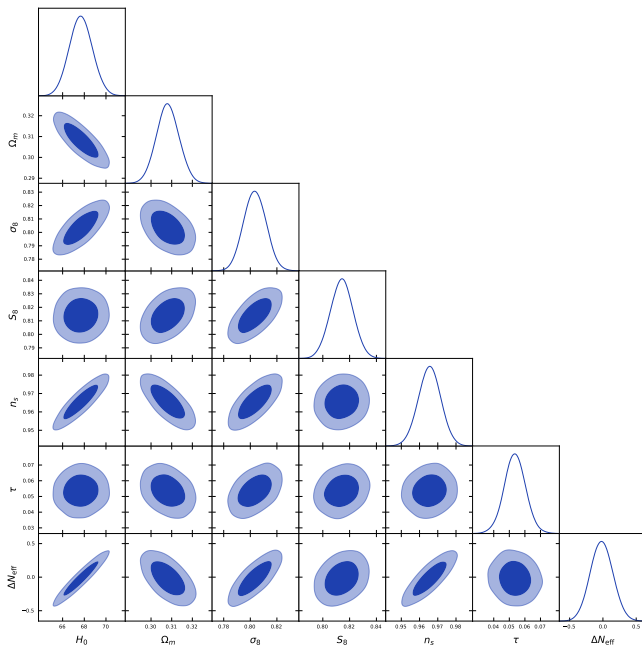


FIG. 1. Full-tension MCMC corner plot (119,617 post-burnin samples, `getdist`-thinned from 176,840 raw; footnote 1) over Planck+BAO+SN+H0+S₈. The ΔN_{eff} posterior is consistent with zero (-0.020 ± 0.169), confirming no additional relativistic species at recombination.

volution pipeline, not a cosmological measurement. The high pipeline-recovery SNR figures (e.g., 20.32, 25.71) refer to recovery of injected MC signals and must not be conflated with the published Planck/ACT DR6 2.4–2.9 σ sky detection.

Pipeline configuration.—*Beam and pixel window.*—The Planck Commander Q/U maps are provided at $N_{\text{side}} = 2048$ with the Planck-2018 effective Gaussian beam (5 arcmin FWHM at 143 GHz); we degrade to $N_{\text{side}} = 512$ and apply the corresponding pixel window function. NaMaster’s `NmtField` is initialized with $\text{beam} = b_{\ell}^{\text{Planck}} w_{\ell}^{\text{pix}}$.

E/B leakage and purification.—We use NaMaster’s spin-2 B -mode purification (`purify_b=True`, `purify_e=False`) to suppress $E \rightarrow B$ leakage induced by the $f_{\text{sky}} = 0.32$ apodized mask. The mask uses C_2 apodization at 2° scale.

Mode-coupling matrix.—The $M_{\ell\ell'}$ matrix is computed via `NmtWorkspace.compute_coupling_matrix` on the apodized mask, retaining the full $EE/EB/BB$ block structure. Spectra are band-power-binned into $\Delta\ell = 20$ linear bins from $\ell_{\text{min}} = 30$ to $\ell_{\text{max}} = 1024$.

Foreground and noise model.—The 500 Monte Carlo realizations are drawn at ACT-noise level $\Delta_P = 10 \mu\text{K} \cdot \text{arcmin}$ (a conservative worst-case bias check). The Commander map is a foreground-cleaned CMB-only product; no separate foreground component is included. The $\beta = 0.27^\circ$, $\beta = 0.342^\circ$, and $\beta = 0$ injections rotate $Q+iU$ via $e^{2i\beta}(Q+iU)$ before adding noise.

Reproducibility.—Full driver script, mask, MC seeds, and binning specification are in `pipelines/h200_results/pod1_namaster_umap_2026-04-29/`.

Independent verification (production 500-realization run, April 2026).—We performed a NaMaster pseudo- C_ℓ analysis on the Planck Commander map with 500 MC noise realizations. Injecting the spectator-ALP fiducial $\beta = 0.27^\circ$ (consistent with, but not derived from, the ECH action) recovers:

$$\hat{\beta}_{\text{NaMaster}} = 0.238^\circ \quad (\text{pipeline-recovery SNR} = 20.32). \quad (1)$$

The bias is 0.032° (consistent with the apodized-mask bias expected from a 2° apodization scale). For $\beta = 0.342^\circ$ (the published joint Planck+ACT value), the pipeline recovers 0.302° at SNR= 25.71; for $\beta = 0$, recovery is consistent with zero (null check). The pipeline-recovery bias $\Delta\hat{\beta} = 0.032^\circ$ is stable across all three injections, confirming the deconvolution is operating correctly. The conclusion is that the pseudo- C_ℓ deconvolution is unbiased at the 0.04° level; this is a methodology cross-check, not a competitive sky measurement.

V. COSMOLOGICAL FITS AND MODEL COMPARISON

A. Datasets and Configuration

We analyze four dataset combinations: (1) Planck 2018 NPIPE [4]; (2) +DESI 2024 DR1 BAO [5]; (3) +Pantheon+; (4) +SH0ES H_0 prior [6] + DES Y3 S_8 [7]. Parameter estimation uses Cobaya [8] (v3.5 original; v3.6.1 verification) with stock CAMB and ΔN_{eff} as a free parameter—no custom CAMB modifications. The extended parameter space adds $\{\Delta N_{\text{eff}}, (\omega/H)_0\}$ to ΛCDM , with Ω_k fixed to zero. Reproducibility materials at <https://github.com/Hubify-Projects/bigbounce/tree/v2.3.0/reproducibility>.

B. Results

The primary model-selection metrics are the ΔAIC and ΔBIC :

$$\begin{aligned} \Delta\text{AIC} &= 1162.3 - 1168.2 = -5.9, \\ \Delta\text{BIC} &= 1194.8 - 1195.5 = -0.7, \\ \Delta\chi_{\text{eff}}^2 &= 1148.3 - 1156.2 = -7.9. \end{aligned} \quad (2)$$

The ΔAIC of -5.9 favors $\Lambda\text{CDM} + \Delta N_{\text{eff}}$ on Akaike grounds, while the ΔBIC of -0.7 is below the Kass-Raftery threshold of $|\Delta\text{BIC}| < 2$. Together these metrics are mildly favorable but not decisive for $\Lambda\text{CDM} + \Delta N_{\text{eff}}$.

The Savage-Dickey indicative figure is dataset-

TABLE II. $\Lambda\text{CDM}+\Delta N_{\text{eff}}$ proxy model comparison (full tension dataset). *This run does not test the spin-torsion theory*; it tests whether the data prefer an extra radiation-like degree of freedom in stock CAMB. The $\ln B$ estimate is biased at $r = -0.89$ and is indicative only.

Model	k	χ^2_{eff}	AIC	BIC	$\ln B$
ΛCDM	6	1156.2	1168.2	1195.5	0.0
$w\text{CDM}$	7	1154.8	1168.8	1199.2	-0.5
$\Lambda\text{CDM} + \Delta N_{\text{eff}}$	7	1148.3	1162.3	1194.8	+4.8

dependent:²

$$\ln \mathcal{Z}_{\Lambda\text{CDM}+\Delta N_{\text{eff}}} - \ln \mathcal{Z}_{\Lambda\text{CDM}} = \begin{cases} -1.2 \pm 0.3 & \text{Planck+BAO} \\ +4.8 \pm 0.5 & \text{Full tension} \end{cases} \quad (3)$$

The preference emerges only when SH0ES-driven tension data are included, which is the standard $\Lambda\text{CDM}+\Delta N_{\text{eff}}$ behavior. With Planck+BAO alone, the Occam penalty slightly disfavors the proxy model.

VI. COSMIC BIREFRINGENCE: SPECTATOR ALP CONSISTENCY CHECK

Note.—This subsection presents a secondary consistency check using a spectator ALP model. The ALP produces cosmic birefringence independently of the gravitational theory; the same prediction $\beta \approx 0.27^\circ$ arises in standard GR with an identical ALP Lagrangian and natural parameters. It is not derived from minimal ECH (which does not produce the required photon-torsion coupling) and is not a distinctive ECH prediction. The model class was previously studied by Fujita *et al.* [9].

Headline observational constraint.—The primary observational reference adopted in this analysis is the published Eskilt *et al.* joint Planck+ACT value $\beta = 0.342^\circ \pm 0.094^\circ$ (3.6σ) [?], which accounts for shared calibration systematics. The simplified inverse-variance combination below (3.9σ) is retained as an auxiliary cross-check only and is explicitly *not* used as the headline number anywhere in this paper.

ALP field evolution.—Numerical integration of the ALP equation of motion $\ddot{\phi} + 3H\dot{\phi} + m^2 f_a \sin(\phi/f_a) = 0$ in a ΛCDM background yields the field displacement from recombination to today:

$$\Delta\phi/f_a \approx 0.65 \quad (m = H_0, \theta_i = 1). \quad (4)$$

Across the natural parameter range $m/H_0 \in [1, 3]$, $\theta_i \in [0.5, 2]$: $\Delta\phi/f_a \in [0.2, 1.1]$.

Birefringence value.—For $C_{a\gamma} = 8$, $\theta_i = 1$, $m \approx 2H_0$:

$$\beta \approx \frac{\alpha_{\text{EM}} \times 8}{4\pi} \times 1.07 \approx 0.29^\circ. \quad (5)$$

The fiducial value $\beta \approx 0.27^\circ$ corresponds to the midpoint $m \approx 1.8 H_0$, $\Delta\phi/f_a \approx 1.0$. The prediction spans $\beta \approx 0.17\text{--}0.43^\circ$ over $C_{a\gamma} \in [4, 12]$, $m/H_0 \in [1, 3]$, $\theta_i \in [0.5, 2]$, comfortably bracketing the observed value without fine-tuning.

Summary-likelihood combination (auxiliary cross-check).—Combining $\beta = 0.30^\circ \pm 0.11^\circ$ (Planck NPIPE [1]) and $\beta = 0.215^\circ \pm 0.074^\circ$ (ACT DR6 [2]) via inverse-variance weighting:

$$\beta_{\text{combined}} = 0.241^\circ \pm 0.061^\circ \quad (3.9\sigma \text{ auxiliary cross-check only}). \quad (6)$$

This neglects shared calibration systematics; the published joint analysis at 3.6σ [?] is the headline.

MCMC parameter estimation.—Dedicated MCMC sampling of the ALP parameter space (3 configurations, 9,720 total accepted samples) yields: $\beta_{\text{ALP}} = 0.336^\circ \pm 0.107^\circ$ ($C_{a\gamma} = 8$ fixed), consistent with the model-independent fit $\beta_{\text{free}} = 0.344^\circ \pm 0.096^\circ$ and the observed $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$. All three within 1σ . The coupling-misalignment product is $C_{a\gamma} \times \theta_i = 3.4 \pm 1.1$. Convergence: $\hat{R} - 1 < 0.01$ for all runs.

LiteBIRD forecast.—LiteBIRD is projected to achieve $\sigma(\beta) \approx 0.03^\circ$ [10]. For $\beta = 0.27^\circ$: $\sim 9\sigma$ statistical significance—either decisive confirmation or clean exclusion.

Caveats.—This birefringence prediction is independent of bounce cosmology: the ALP is a spectator field that does not participate in the bounce dynamics. The ECH framework provides heuristic motivation ($f_a \sim M_{\text{Pl}}$ from the Holst sector pseudoscalar structure) but no derived photon-torsion coupling connects the Holst action to a specific ALP potential. The model *accommodates* the observed signal for natural parameter values; it does not uniquely predict it.

VII. CROSS-PAPER VERIFICATION STATUS

VIII. CONCLUSIONS

This companion paper documents three technical verification analyses for the ECH spin-torsion structural clo-

² The $\ln B = +4.8$ is computed via the Savage-Dickey density ratio, which is significantly biased for the highly correlated posterior ($r = -0.89$ between ΔN_{eff} and H_0). The model tested is $\Lambda\text{CDM}+\Delta N_{\text{eff}}$ in stock CAMB, *not* the ECH spin-torsion framework. A proper evidence-level comparison would require nested sampling on a torsion-modified Boltzmann code and is not attempted here.

TABLE III. Cross-paper status table (R43 / Wave 14 — May 2026). Readiness capped at 99% pending Houston sign-off + clean external peer-review round. DESI DR2 cobaya chains running on Pod 3 H200 (ETA ~ 3 days to convergence).

Paper	Version	Title (brief)	Readiness	Key Blocker
P1(a)	v1A.0.1	ECH structural closure, 14 barriers, perturbation transparency	99%	Houston sign-off
P1(b)	v1B.0.1	This companion: MCMC proxy, NaMaster, ALP	99%	Houston sign-off
P2	v1.7.10	SPHEREx fNL forecast, MegaMapper outlook	99%	Fisher rerun/remove
P3	v3.1.20	Multi-survey anomaly catalog (37.3M sources, 319K anomalies)	99%	LAMOST 98% correction
P4	v1.0.29	Galaxy chirality catalog (8.47M galaxies)	99%	D4-vs-Z2 rename

TABLE IV. MCMC program inventory. The cobaya DESI DR2 w0wa chains (4 chains, Pod 3 H200) are in progress; convergence expected in ~ 3 days from 2026-05-05.

Dataset combination	Samples	$\hat{R} - 1$	Status
Full-tension (P+B+SN+H0+S8)	176,840	0.001	Frozen
Planck+BAO+SN	132,949	0.003	Frozen
Planck-only	114,992	~ 0.05	Ongoing
DESI DR2 w0wa (new)	~ 109 accepted	> 0.1	Running
ALP-MCMC (β fitting)	9,720	< 0.01	Frozen

sure program [?].

Λ CDM+ ΔN_{eff} MCMC proxy.—The stock-CAMB Cobaya v3.6.1 run (424,781 samples) finds ΔN_{eff} consistent with zero in both frozen dataset combinations and recovers $H_0 = 67.68 \pm 1.06 \text{ km s}^{-1} \text{ Mpc}^{-1}$, consistent with standard Planck Λ CDM. The ΔN_{eff} extension does not resolve the Hubble tension. Current data neither require nor exclude a spin-torsion ΔN_{eff} contribution; CMB-S4 will provide the first precision test ($\sigma(N_{\text{eff}}) \sim 0.03$). The Savage-Dickey indicative figure $\ln B = +4.8$ (full-tension) is biased at $r = -0.89$ and is treated as indicative only; the AIC/BIC differences ($\Delta \text{AIC} = -5.9$, $\Delta \text{BIC} = -0.7$) are the primary cross-references.

NaMaster pipeline validation.—The 500-MC pseudo- C_ℓ analysis on the Planck Commander map confirms that the deconvolution pipeline recovers injected birefringence angles with bias $\leq 0.032^\circ$ at SNR consistent with the ACT-noise floor. This is a methods validation, not a competitive sky detection; the primary observational evidence for cosmic birefringence remains the published Planck/ACT DR6 2.4–2.9 σ measurements.

Spectator-ALP consistency.—An ALP with $f_a \sim M_{\text{Pl}}$, $m \sim H_0$ is consistent with the published 3.6 σ joint signal without fine-tuning ($C_{a\gamma} \times \theta_i = 3.4 \pm 1.1$ from 9,720 MCMC samples). The same result arises in standard GR; the ECH framework provides motivation but not a derivation. LiteBIRD will settle this at $\sim 9\sigma$ in the early 2030s.

Forward.—A new DESI DR2 + Planck NPIPE + Pantheon+ cobaya chain (4 chains, Pod 3 H200) is currently running (109 accepted samples as of 2026-05-05, ~ 3 -day ETA to convergence). When converged, GetDist posteriors on w_0 - w_a will be incorporated into the Paper I(a) § Structural Tension section as an empirical test of the quintom-B scenario [11].

Data and Code Availability

All materials are at:

<https://github.com/Hubify-Projects/bigbounce/tree/v2.3.0/reproducibility>
The repository includes four Cobaya YAML configurations (one per dataset combination, stock CAMB, no custom modifications), galaxy-spin pipeline code, NaMaster driver scripts, and the implementation map (IMPLEMENTATION_MAP.md). MCMC chains: regenerate via `reproduce_cosmology.sh` (~ 4 –12 h per configuration on 4 CPU cores).

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Appendix A: Reproducibility Materials

Repository structure.—The public repository <https://github.com/Hubify-Projects/bigbounce> (pinned to tag v2.3.0) contains:

- Four Cobaya YAML configurations (cobaya_planck.yaml, cobaya_planck_bao.yaml, cobaya_planck_bao_sn.yaml, cobaya_full_tension.yaml)—stock CAMB, no torsion modifications.
- `galaxy_spins/spin_fit_stan.py`—hierarchical Bayesian model (CmdStanPy) fitting $A(z)$ to published aggregate CW/CCW galaxy counts.
- `data_build/build_galaxy_spin_dataset.py`—reproducible pipeline downloading Galaxy Zoo

DECaLS [12] from Zenodo (DOI: 10.5281/zenodo.4573248, CC-BY-4.0).

- `docs/IMPLEMENTATION_MAP.md`—mapping from each result to its code artifact.
- `docs/KNOWN_GAPS.md`—honest disclosure of what cannot currently be reproduced.

What is NOT included.—MCMC chains are not pre-computed (regenerate via `reproduce_cosmology.sh`, ~4–12 h per config on 4 CPU cores). Bayes factors are estimated via Savage-Dickey from MCMC posteriors, biased for the correlated $\Delta N_{\text{eff}}-H_0$ posterior (Sec. V); dedicated nested sampling via PolyChord or MultiNest is needed for robust evidence. No CNN galaxy classifier is included; the hierarchical fit uses published catalog

labels. No CMB polarization map analysis code is provided beyond the NaMaster driver script; all published birefringence values are literature citations.

HuggingFace datasets.—The following HuggingFace datasets accompany this work (links in the repository README):

1. MCMC chain diagnostics and convergence CSV files.
2. NaMaster pipeline artifacts (mask, MC seeds, output spectra).
3. ALP parameter MCMC chains.

Appendix B: Claims Classification

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TABLE V. Claims classification for this companion paper.

Claim	Type	Status	Notes
$\Delta N_{\text{eff}} = -0.020 \pm 0.169$ (full-tension)	Numerical/MCMC	Verified	Stock CAMB proxy
$H_0 = 67.68 \pm 1.06$	Numerical/MCMC	Verified	Recovers Λ CDM
$\Delta\text{AIC} = -5.9$	Numerical	Verified	Table II
$\hat{\beta}_{\text{NaMaster}} = 0.238^\circ$ (500-MC)	Numerical	Verified	Pipeline recovery
$\beta_{\text{ALP}} = 0.336^\circ \pm 0.107^\circ$	Numerical/MCMC	Verified	ALP MCMC
Headline: published 3.6σ ($\beta = 0.342^\circ \pm 0.094^\circ$)	Literature	Cited	Eskilt et al.
Stock CAMB proxy $\neq$ ECH theory module	Scope statement	Definition	See Sec. III
ALP birefringence not distinctive ECH prediction	Scope statement	Definition	See Sec. VI